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END-TERM REPORT

# **Decoding Digital Payments in India: UPI's Business Model and the Emergence of CBDC**

*An Analyst Assessment of CBDC's Implications for India's Digital Payments Ecosystem*

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# Executive Summary

India's Unified Payments Interface (UPI) has become the default rail for retail digital payments, clearing 23.2 billion transactions worth ₹29.9 lakh crore in May 2026 alone and accounting for roughly 81-85% of retail digital payment volume in the country. It achieved this scale while operating under a zero Merchant Discount Rate (MDR) regime, meaning the payment rail itself generates negligible direct revenue for the banks and payment service providers (PSPs) - such as PhonePe, Google Pay, and Paytm - that built the consumer-facing apps on top of it. The government currently plugs this gap with a budgetary incentive scheme, but that scheme's growth has not kept pace with UPI's own transaction growth, and Budget 2026 cut the allocation even as volumes rose - reopening a sustainability debate that has simmered since 2020.

Layered onto this is the Reserve Bank of India's Central Bank Digital Currency (CBDC), the e₹, which is also structured as a no-MDR, RBI-issued instrument. In theory, a free, RBI-backed digital currency competing for the same use cases could squeeze PSPs further, or intensify the argument that UPI's free-payments model is unsustainable. In practice, this report finds that e₹'s adoption curve, at a comparable stage in its life cycle, sits several orders of magnitude behind UPI's own early trajectory - suggesting the disintermediation and sustainability risks CBDC poses to the existing ecosystem are real in principle but distant in practice, likely a late-2020s-into-2030s consideration rather than an immediate threat.

## 1. The Zero-MDR Debate: Who Pays for Free UPI Today?

Merchant Discount Rate (MDR) is the fee a bank charges a merchant to process a digital payment - the standard mechanism by which card networks and payment rails monetize transactions. Since January 2020, MDR on RuPay debit cards and BHIM-UPI transactions has been set to zero by regulatory fiat, through amendments to the Payment and Settlement Systems Act, 2007 and the Income-tax Act, 1961. The explicit goal was to remove any friction to adoption during UPI's critical early-growth years.

Zero MDR does not mean zero cost. Processing, settlement, fraud monitoring, and customer-service infrastructure still have to be paid for. To compensate acquiring banks (and, downstream, issuer banks and PSPs), the Centre runs a budgetary incentive scheme that reimburses a small percentage - currently 0.15% - of the value of low-ticket UPI person-to-merchant (P2M) transactions made by small merchants.

| Financial Year | Government incentive outlay                            | UPI transaction growth (approx.)                |
|----------------|--------------------------------------------------------|-------------------------------------------------|
| FY 2021-22     | ₹1,389 crore                                           | BHIM-UPI volume up ~106% YoY                    |
| FY 2022-23     | ₹2,210 crore                                           | Continued triple-digit growth off a larger base |
| FY 2023-24     | ₹3,631 crore (incl. RuPay)                             | ~131 billion annual transactions                |
| FY 2024-25     | ₹1,923 crore (BHIM-UPI component)                      | ~228 billion annual transactions (+~74% YoY)    |
| FY 2025-26     | ₹437 crore budgeted → revised to ₹2,196 crore mid-year | ~241.6 billion annual transactions (+30% YoY)   |

|                          |                                                         |                                                                          |
|--------------------------|---------------------------------------------------------|--------------------------------------------------------------------------|
| FY 2026-27 (Budget 2026) | ₹2,000 crore – a ~9% cut from the FY26 revised estimate | Volumes still climbing toward NPCI's 1-billion-transactions-a-day target |
|--------------------------|---------------------------------------------------------|--------------------------------------------------------------------------|

The pattern in Table 1 is the crux of the sustainability debate: transaction volumes have grown at 30-100%+ year-on-year in recent years, while the incentive outlay has grown far more erratically and, in the FY 2026-27 Budget, was actually cut by roughly 9% relative to the prior year's revised estimate - even though daily transaction counts kept climbing toward 30 crore a day. Industry body Payments Council of India (PCI) publicly said it had hoped for an allocation above ₹10,000 crore and warned that funding growth purely through unpredictable annual budgetary allocations "will choke the entire ecosystem." A Parliamentary Standing Committee on Finance report in early 2026 went further, with the Department of Financial Services itself submitting that the absence of MDR makes the UPI ecosystem financially unsustainable, and floating the goal that UPI should become self-reliant rather than dependent on a ₹2,000 crore annual line item.

PCI's proposed fix is narrowly targeted rather than a blanket reversal: introduce a modest MDR (around 0.3%) on P2M transactions only for larger merchants (annual turnover above ₹20 lakh), while keeping P2P transfers and small-merchant payments free. This reframes the debate away from "should UPI charge users" (politically untouchable) toward "should large, well-capitalised merchants absorb a small processing cost" (a more defensible ask). Whether this happens is uncertain and politically sensitive - free UPI has become a point of national pride and financial-inclusion messaging - but the direction of travel in policy circles by mid-2026 is toward acknowledging that indefinite, fully-subsidised zero-MDR is not a stable long-run equilibrium.

## 2. PSP Business Models: Where the Real Money Is Made

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If the payment rail itself is close to a break-even public good, how do PhonePe, Google Pay, and Paytm sustain businesses valued in the billions of dollars? The answer is that UPI functions as a customer-acquisition and distribution channel, not a revenue line. Payments bring hundreds of millions of users and tens of millions of merchants into an app; the actual monetization happens one layer up, in financial products and merchant services layered on top.

PhonePe's disclosures around its 2025-26 IPO filing are illustrative of the industry pattern. Revenue from Financial Services - largely commissions on loan origination and insurance distribution, since PhonePe holds neither an NBFC nor an insurer license - grew from just ₹28 crore in FY23 to ₹452.8 crore in H1 FY26, and its share of operating revenue expanded from under 1% to 11.55% over the same window. Merchant Payments revenue (subscriptions and services tied to its device network of roughly 92 lakh deployed EDC machines and Smart speakers) grew from 14.75% to 30.78% of revenue. Together, these two lines - neither of which is UPI transaction-fee income - now account for 42% of PhonePe's total revenue. As of September 2025 the company had facilitated ₹14,270 crore in loans through 56 lending partners and sold 1.85 crore insurance policies across 29 insurers, all on a pure distribution/commission basis. The company was still loss-making (₹1,727 crore net loss in FY25), but losses are narrowing as this financial-services layer scales.

Google Pay and Paytm follow the same broad logic with different execution. Google Pay has struggled to build a comparably deep financial-services stack and leans more on the strategic value of payments data and distribution within the broader Google ecosystem. Paytm anchors merchant relationships through its large

Sound box device base (over 1.17 crore units) - not because the devices themselves are profitable, but because they create the merchant touchpoint from which lending, insurance, and cash-management products are cross-sold. At the infrastructure layer, NPCI itself - which operates UPI as a not-for-profit utility - posted a surplus of only ₹1,552 crore in FY2025 on a network that processed roughly \$3 trillion of value, underscoring just how thin the margins are at the rail level relative to the volumes involved.

The structural takeaway: PSP profitability is decoupled from UPI transaction economics. The apps monetize the relationship, not the rail. This matters directly for assessing CBDC's threat, addressed next.

### 3. CBDC Meets the PSP Model: Threat or Non-Event?

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A natural fear for PSPs is that a central-bank-issued digital currency could disintermediate them entirely - RBI could, in theory, distribute the e₹ directly to citizens' wallets and cut PhonePe, Google Pay, and Paytm out of the loop altogether, taking both the transaction relationship and the associated data with it. The evidence from how RBI has actually architected the e₹ suggests this is not the design intent, at least for now.

India's CBDC follows a two-tier, intermediated model deliberately built to avoid disintermediating existing PSPs: RBI issues the digital rupee, but distribution and end-user interaction happen through banks and authorised intermediaries, explicitly including the payment ecosystem already built around UPI. In a concrete sign of convergence rather than competition, the State Bank of India has already implemented UPI interoperability within its e₹ app, letting CBDC-wallet users scan ordinary UPI merchant QR codes. Rather than building a rival acceptance network from scratch, the e₹ is, so far, riding on UPI's existing merchant-QR infrastructure. Multilateral research (BIS, IMF) on CBDC design broadly echoes this: intermediated architectures preserve a role for PSPs in servicing, onboarding, and increasingly in monetising CBDC transaction and identity data for products like credit scoring - potentially a new revenue surface for PSPs rather than a threat to the existing one.

Where CBDC does bite conceptually is on the no-MDR-by-design point raised in the assignment brief. The e₹ has no merchant discount rate at all - it is designed to behave like digital cash, a direct RBI liability rather than a bank-intermediated transfer. This means CBDC does not "solve" UPI's sustainability problem by offering a monetisable alternative; if anything, it adds a second permanently-free, publicly-backed payment instrument to the landscape, which makes it politically harder, not easier, to introduce even a narrow MDR on UPI later - a large merchant could reasonably ask why it should pay 0.3% on a UPI transaction when an RBI-issued digital rupee transaction, once it scales, would cost nothing. In that sense, CBDC's design intensifies the zero-MDR debate rhetorically even though, as the next section shows, it does not yet intensify it materially, because CBDC volumes remain negligible.

The more tangible near-term interaction is in a narrow but growing niche: government-to-citizen welfare disbursement. RBI has begun routing parts of India's roughly \$80 billion welfare system through the e₹ in about ten pilot programmes - food subsidies via programmable CBDC in Gujarat, Puducherry, and Chandigarh, redeemable only at fair-price shops and identified merchants. This is a genuine example of CBDC bypassing PSP apps for a specific transaction category, using programmability (spending restricted to eligible commodities) that UPI cannot natively replicate. But it is a targeted government-payments use case, not a substitute for everyday P2P or P2M retail payments where PSPs currently earn their distribution value.

## 4. Adoption Trajectories: e₹ versus UPI's Early Years

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The clearest way to judge how near or distant CBDC's implications really are is to line up its adoption curve against UPI's own early trajectory, stage for stage, rather than against UPI's current (mature) scale.

| Time since launch   | UPI (launched April 2016)                                                         | e₹-Retail / CBDC (launched Dec 2022)                                                                        |
|---------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Year 1              | ~17.9 million transactions in FY 2016-17; 21 banks live                           | Pilot phase; single-digit lakh users, closed user group                                                     |
| ~1.5-2 years        | ~920 million transactions in FY 2017-18 (UPI 2.0 launched Aug 2018)               | ~1 million transactions/day target reached (Dec 2023)                                                       |
| ~3-3.5 years        | Crossed 1 billion transactions in a single month for the first time (Oct 2019)    | ~120 million cumulative transactions; ~8 million users (late 2025)                                          |
| ~3.5 years+         | Second billion/month reached in ~1 more year - inflection into exponential growth | ~175 million cumulative transactions; ~12 million users; e₹ in circulation ↓ 24% to ₹771.7 crore (mid-2026) |
| 2026 scale (latest) | ~23.2 billion transactions in a single month (May 2026), worth ₹29.9 lakh crore   | ~175 million transactions in total since launch - roughly a single day of UPI volume                        |

The comparison is stark. UPI took roughly three-and-a-half years to go from launch to clearing one billion transactions in a single month, and from there the growth became genuinely exponential - it now clears that same one-billion mark roughly every day. The e₹, at a similar distance from its own December 2022 launch (i.e., by mid-2026), has processed only about 175 million transactions in total across its entire lifetime - less than a single day's UPI volume today, and a small fraction of even UPI's first-month-of-year-three run rate. e₹ user count (roughly 12 million) is also a small fraction of UPI's 500-million-plus user base. Perhaps most tellingly, the value of e₹ actually in circulation fell 24% year-on-year to ₹771.7 crore as of March 2026, even as RBI expanded the number of active pilots - suggesting that where CBDC transaction counts are rising, it is substantially because of targeted government pushes (the welfare DBT pilots) rather than organic, UPI-style consumer pull.

This gap is best explained by the different problems the two instruments solve. UPI arrived into a market with no fast, free, interoperable digital payment rail - it filled a genuine infrastructure gap, and its zero-MDR design removed the last friction to adoption at exactly the moment smartphone and data penetration made mass uptake possible. The e₹, by contrast, is entering a market where UPI already does everything a retail CBDC would do from a user's point of view - instant, free, interoperable, universally accepted - and offers no obvious incremental convenience to an ordinary user or merchant. Its most distinctive features (offline capability, programmability, direct RBI-liability status) are meaningful to specific institutional and policy use cases - wholesale settlement, welfare disbursement, cross-border trade - but are largely invisible to a shopkeeper choosing between an e₹ QR code and a UPI QR code that already works everywhere.

## 5. Synthesis: How Near or Distant Is the Threat?

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Bringing the three strands together points to a layered, time-differentiated assessment rather than a single verdict.

- Near-term (2026-2028): Negligible direct threat to PSP business models. e₹ volumes are roughly four orders of magnitude smaller than UPI's, PSP revenue already comes overwhelmingly from lending, insurance, and merchant services rather than payment-transaction economics, and RBI's intermediated CBDC architecture explicitly preserves a role for existing PSPs and banks rather than disintermediating them. The zero-MDR sustainability debate in this window is being driven almost entirely by UPI's own scale (30 crore transactions a day against a shrinking incentive budget), not by CBDC.
- Medium-term (2028-2032): The interaction becomes more consequential but is still an amplifier of the existing sustainability debate rather than an independent threat. If e₹ achieves even a modest share of retail volume - most plausibly first in welfare disbursement, government payments, and wholesale/interbank settlement via the Unified Markets Interface - its permanently no-MDR design will make it politically harder to introduce any MDR on UPI, entrenching the current subsidy model rather than resolving it. CBDC transaction and identity data may also open a genuinely new monetisation surface for PSPs (credit scoring, targeted financial products), which cuts against the disintermediation narrative and toward a story of PSPs absorbing CBDC as one more rail they service, much as they already service UPI, cards, and wallets.
- Long-term (2032+): This is where a structural threat becomes plausible in principle - if programmable, RBI-issued money becomes a significant share of low-value transactions, as some industry roadmaps envisage, the case for maintaining fee-generating intermediaries in the payment chain weakens. But this horizon is speculative relative to 2026 evidence and depends on policy choices (interest-bearing status, holding limits, merchant incentives) that RBI has not yet made.

Overall, this report's assessment is that CBDC's implications for India's digital payments ecosystem are real but currently distant, and that the zero-MDR sustainability question is, for the next several years, a UPI problem far more than a CBDC problem. The government's own budgetary math - cutting the UPI incentive allocation even as UPI volumes surge 30% a year - is the more urgent and immediate fault line in India's digital payments model; e₹, at its current scale, is a policy experiment worth watching rather than a competitive force actively reshaping PSP economics today.

## Conclusion

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UPI's zero-MDR design solved India's adoption problem but created a standing subsidy dependency that budget allocations have started to strain against, even as PSPs have adapted by monetising the customer relationship - through lending, insurance, and merchant services - rather than the payment itself. CBDC enters this picture as a second no-MDR, RBI-backed instrument, which in principle could either relieve pressure on PSPs (by giving RBI a policy tool that keeps payments free without further burdening the Union budget) or add to it (by making any future reintroduction of MDR politically harder). At its current adoption trajectory - orders of magnitude behind where UPI stood at the same age - the e₹ is not yet large enough to force that question. The more immediate and consequential story for India's digital payments ecosystem in 2026 is the widening gap between UPI's transaction growth and the government's willingness to keep fully funding it for free.

## Key Sources

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